

Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.46)

Metric

$$ds^2 = \left(\frac{h-1}{h+1} \right) dt \otimes dt + \left(\frac{16(h^4 + 4h^3 + 6h^2 + 4h + 1)e^g}{R^4 r^4 + 8R^2 r^2 + 16} \right) dr \otimes dr + \left(\frac{16(r^2 h^4 + 4r^2 h^3 + 6r^2 h^2 + 4r^2 h + r^2)e^g}{R^4 r^4 + 8R^2 r^2 + 16} \right) d\theta \otimes d\theta + \left(\frac{16(r^2 h^4 + 4r^2 h^3 + 6r^2 h^2 + 4r^2 h + r^2)e^g \sin(\theta)^2}{R^4 r^4 + 8R^2 r^2 + 16} \right) d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$h = \frac{1}{4} \sqrt{R^2 r^2 + 4} m r e^{(-\frac{1}{2}g)}$$

$$g(t) = g$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \sqrt{-\frac{h}{h+1} + \frac{1}{h+1}} \right) dt + \left(-4 \sqrt{\frac{1}{2}} \sqrt{\frac{h^4}{R^4 r^4 + 8R^2 r^2 + 16} + \frac{4h^3}{R^4 r^4 + 8R^2 r^2 + 16} + \frac{6h^2}{R^4 r^4 + 8R^2 r^2 + 16} + \frac{4h}{R^4 r^4 + 8R^2 r^2 + 16} + \frac{1}{R^4 r^4 + 8R^2 r^2 + 16}} e^{(\frac{1}{2}g)} \right) dr$$

$$n = \left(\sqrt{\frac{1}{2}} \sqrt{-\frac{h}{h+1} + \frac{1}{h+1}} \right) dt + \left(4 \sqrt{\frac{1}{2}} \sqrt{\frac{h^4}{R^4 r^4 + 8R^2 r^2 + 16} + \frac{4h^3}{R^4 r^4 + 8R^2 r^2 + 16} + \frac{6h^2}{R^4 r^4 + 8R^2 r^2 + 16} + \frac{4h}{R^4 r^4 + 8R^2 r^2 + 16} + \frac{1}{R^4 r^4 + 8R^2 r^2 + 16}} e^{(\frac{1}{2}g)} \right) dr$$

$$m = \left(4 \sqrt{\frac{1}{2}} \sqrt{\frac{h^4}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{4 h^3}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{6 h^2}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{4 h}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{1}{R^4 r^4 + 8 R^2 r^2 + 16}} r e^{\left(\frac{1}{2} g\right)} \right) d\theta \\ + \left(4i \sqrt{\frac{1}{2}} \sqrt{\frac{h^4}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{4 h^3}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{6 h^2}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{4 h}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{1}{R^4 r^4 + 8 R^2 r^2 + 16}} r e^{\left(\frac{1}{2} g\right)} \sin(\theta) \right) d\phi$$

$$\bar{m} = \left(4 \sqrt{\frac{1}{2}} \sqrt{\frac{h^4}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{4 h^3}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{6 h^2}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{4 h}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{1}{R^4 r^4 + 8 R^2 r^2 + 16}} r e^{\left(\frac{1}{2} g\right)} \right) d\theta \\ + \left(-4i \sqrt{\frac{1}{2}} \sqrt{\frac{h^4}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{4 h^3}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{6 h^2}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{4 h}{R^4 r^4 + 8 R^2 r^2 + 16} + \frac{1}{R^4 r^4 + 8 R^2 r^2 + 16}} r e^{\left(\frac{1}{2} g\right)} \sin(\theta) \right) d\phi$$

Contravariant components

$$l = \left(\frac{\sqrt{h+1}\sqrt{-h+1}}{\sqrt{2}h - \sqrt{2}} \right) dt + \left(-\frac{R^2 r^2 e^{\left(-\frac{1}{2} g\right)}}{4(\sqrt{2}h^2 + 2\sqrt{2}h + \sqrt{2})} - \frac{e^{\left(-\frac{1}{2} g\right)}}{\sqrt{2}h^2 + 2\sqrt{2}h + \sqrt{2}} \right) dr$$

$$n = \left(\frac{\sqrt{h+1}\sqrt{-h+1}}{\sqrt{2}h - \sqrt{2}} \right) dt + \left(\frac{R^2 r^2 e^{\left(-\frac{1}{2} g\right)}}{4(\sqrt{2}h^2 + 2\sqrt{2}h + \sqrt{2})} + \frac{e^{\left(-\frac{1}{2} g\right)}}{\sqrt{2}h^2 + 2\sqrt{2}h + \sqrt{2}} \right) dr$$

$$m = \left(\frac{R^2 r^2 e^{\left(-\frac{1}{2} g\right)}}{4(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r)} + \frac{e^{\left(-\frac{1}{2} g\right)}}{\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r} \right) d\theta + \left(\frac{i R^2 r^2 e^{\left(-\frac{1}{2} g\right)}}{4(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r) \sin(\theta)} + \frac{i e^{\left(-\frac{1}{2} g\right)}}{(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r) \sin(\theta)} \right) d\phi$$

$$\bar{m} = \left(\frac{R^2 r^2 e^{\left(-\frac{1}{2} g\right)}}{4(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r)} + \frac{e^{\left(-\frac{1}{2} g\right)}}{\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r} \right) d\theta + \left(-\frac{i R^2 r^2 e^{\left(-\frac{1}{2} g\right)}}{4(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r) \sin(\theta)} - \frac{i e^{\left(-\frac{1}{2} g\right)}}{(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r) \sin(\theta)} \right) d\phi$$

Directional Derivatives

$$D = \frac{\sqrt{h+1}\sqrt{-h+1}}{\sqrt{2}h - \sqrt{2}} \partial_t - \frac{R^2 r^2 e^{\left(-\frac{1}{2} g\right)}}{4(\sqrt{2}h^2 + 2\sqrt{2}h + \sqrt{2})} - \frac{e^{\left(-\frac{1}{2} g\right)}}{\sqrt{2}h^2 + 2\sqrt{2}h + \sqrt{2}} \partial_r$$

$$\Delta = \frac{\sqrt{h+1}\sqrt{-h+1}}{\sqrt{2}h - \sqrt{2}} \partial_t + \frac{R^2 r^2 e^{(-\frac{1}{2}g)}}{4(\sqrt{2}h^2 + 2\sqrt{2}h + \sqrt{2})} + \frac{e^{(-\frac{1}{2}g)}}{\sqrt{2}h^2 + 2\sqrt{2}h + \sqrt{2}} \partial_r$$

$$\delta = \frac{R^2 r^2 e^{(-\frac{1}{2}g)}}{4(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r)} + \frac{e^{(-\frac{1}{2}g)}}{\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r} \partial_\theta + \frac{i R^2 r^2 e^{(-\frac{1}{2}g)}}{4(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r) \sin(\theta)} + \frac{i e^{(-\frac{1}{2}g)}}{(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r) \sin(\theta)} \partial_\phi$$

$$\bar{\delta} = \frac{R^2 r^2 e^{(-\frac{1}{2}g)}}{4(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r)} + \frac{e^{(-\frac{1}{2}g)}}{\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r} \partial_\theta - \frac{i R^2 r^2 e^{(-\frac{1}{2}g)}}{4(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r) \sin(\theta)} - \frac{i e^{(-\frac{1}{2}g)}}{(\sqrt{2}rh^2 + 2\sqrt{2}rh + \sqrt{2}r) \sin(\theta)} \partial_\phi$$

Spin Coefficients

$$\begin{aligned} \rho = & \frac{\sqrt{2}R^2 r^3 e^{(-\frac{1}{2}g)} h \frac{\partial}{\partial r} h}{4(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}R^2 r^2 e^{(-\frac{1}{2}g)} h^2}{8(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}R^2 r^3 e^{(-\frac{1}{2}g)} \frac{\partial}{\partial r} h}{4(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h^3 \frac{\partial}{\partial t} g}{4(rh^4 + 2rh^3 - 2rh - r)} + \frac{\sqrt{2}R^2 r^2 e^{(-\frac{1}{2}g)}}{8(rh^4 + 2rh^3 - 2rh - r)} \\ & - \frac{3\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h^2 \frac{\partial}{\partial t} g}{4(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h^2 \frac{\partial}{\partial t} h}{rh^4 + 2rh^3 - 2rh - r} - \frac{3\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h \frac{\partial}{\partial t} g}{4(rh^4 + 2rh^3 - 2rh - r)} + \frac{\sqrt{2}r e^{(-\frac{1}{2}g)} h \frac{\partial}{\partial r} h}{rh^4 + 2rh^3 - 2rh - r} - \frac{2\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h \frac{\partial}{\partial t} h}{rh^4 + 2rh^3 - 2rh - r} \\ & + \frac{\sqrt{2}e^{(-\frac{1}{2}g)} h^2}{2(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}r\sqrt{h+1}\sqrt{-h+1} \frac{\partial}{\partial t} g}{4(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}r e^{(-\frac{1}{2}g)} \frac{\partial}{\partial r} h}{rh^4 + 2rh^3 - 2rh - r} - \frac{\sqrt{2}r\sqrt{h+1}\sqrt{-h+1} \frac{\partial}{\partial t} h}{rh^4 + 2rh^3 - 2rh - r} - \frac{\sqrt{2}e^{(-\frac{1}{2}g)}}{2(rh^4 + 2rh^3 - 2rh - r)} \end{aligned}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\begin{aligned} \mu = & \frac{\sqrt{2}R^2 r^3 e^{(-\frac{1}{2}g)} h \frac{\partial}{\partial r} h}{4(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}R^2 r^2 e^{(-\frac{1}{2}g)} h^2}{8(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}R^2 r^3 e^{(-\frac{1}{2}g)} \frac{\partial}{\partial r} h}{4(rh^4 + 2rh^3 - 2rh - r)} + \frac{\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h^3 \frac{\partial}{\partial t} g}{4(rh^4 + 2rh^3 - 2rh - r)} + \frac{\sqrt{2}R^2 r^2 e^{(-\frac{1}{2}g)}}{8(rh^4 + 2rh^3 - 2rh - r)} \\ & + \frac{3\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h^2 \frac{\partial}{\partial t} g}{4(rh^4 + 2rh^3 - 2rh - r)} + \frac{\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h^2 \frac{\partial}{\partial t} h}{rh^4 + 2rh^3 - 2rh - r} + \frac{3\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h \frac{\partial}{\partial t} g}{4(rh^4 + 2rh^3 - 2rh - r)} + \frac{\sqrt{2}r e^{(-\frac{1}{2}g)} h \frac{\partial}{\partial r} h}{rh^4 + 2rh^3 - 2rh - r} + \frac{2\sqrt{2}r\sqrt{h+1}\sqrt{-h+1}h \frac{\partial}{\partial t} h}{rh^4 + 2rh^3 - 2rh - r} \\ & + \frac{\sqrt{2}e^{(-\frac{1}{2}g)} h^2}{2(rh^4 + 2rh^3 - 2rh - r)} + \frac{\sqrt{2}r\sqrt{h+1}\sqrt{-h+1} \frac{\partial}{\partial t} g}{4(rh^4 + 2rh^3 - 2rh - r)} - \frac{\sqrt{2}r e^{(-\frac{1}{2}g)} \frac{\partial}{\partial r} h}{rh^4 + 2rh^3 - 2rh - r} + \frac{\sqrt{2}r\sqrt{h+1}\sqrt{-h+1} \frac{\partial}{\partial t} h}{rh^4 + 2rh^3 - 2rh - r} - \frac{\sqrt{2}e^{(-\frac{1}{2}g)}}{2(rh^4 + 2rh^3 - 2rh - r)} \end{aligned}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}R^2r^2 \cos(\theta) e^{(-\frac{1}{2}g)}}{16(rh^2 + 2rh + r) \sin(\theta)} - \frac{\sqrt{2} \cos(\theta) e^{(-\frac{1}{2}g)}}{4(rh^2 + 2rh + r) \sin(\theta)}$$

$$\beta = \frac{\sqrt{2}R^2r^2 \cos(\theta) e^{(-\frac{1}{2}g)}}{16(rh^2 + 2rh + r) \sin(\theta)} + \frac{\sqrt{2} \cos(\theta) e^{(-\frac{1}{2}g)}}{4(rh^2 + 2rh + r) \sin(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\begin{aligned} \epsilon = & -\frac{\sqrt{2}R^2r^2 e^{(-\frac{1}{2}g)} \frac{\partial}{\partial r} h}{16(h^4 + 2h^3 - 2h - 1)} + \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h^3 \frac{\partial}{\partial t} g}{8(h^4 + 2h^3 - 2h - 1)} + \frac{3\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h^2 \frac{\partial}{\partial t} g}{8(h^4 + 2h^3 - 2h - 1)} + \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h^2 \frac{\partial}{\partial t} h}{2(h^4 + 2h^3 - 2h - 1)} \\ & + \frac{3\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h \frac{\partial}{\partial t} g}{8(h^4 + 2h^3 - 2h - 1)} + \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h \frac{\partial}{\partial t} h}{h^4 + 2h^3 - 2h - 1} + \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1} \frac{\partial}{\partial t} g}{8(h^4 + 2h^3 - 2h - 1)} - \frac{\sqrt{2}e^{(-\frac{1}{2}g)} \frac{\partial}{\partial r} h}{4(h^4 + 2h^3 - 2h - 1)} + \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1} \frac{\partial}{\partial t} h}{2(h^4 + 2h^3 - 2h - 1)} \end{aligned}$$

$$\begin{aligned} \gamma = & -\frac{\sqrt{2}R^2r^2 e^{(-\frac{1}{2}g)} \frac{\partial}{\partial r} h}{16(h^4 + 2h^3 - 2h - 1)} - \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h^3 \frac{\partial}{\partial t} g}{8(h^4 + 2h^3 - 2h - 1)} - \frac{3\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h^2 \frac{\partial}{\partial t} g}{8(h^4 + 2h^3 - 2h - 1)} - \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h^2 \frac{\partial}{\partial t} h}{2(h^4 + 2h^3 - 2h - 1)} \\ & - \frac{3\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h \frac{\partial}{\partial t} g}{8(h^4 + 2h^3 - 2h - 1)} - \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1}h \frac{\partial}{\partial t} h}{h^4 + 2h^3 - 2h - 1} - \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1} \frac{\partial}{\partial t} g}{8(h^4 + 2h^3 - 2h - 1)} - \frac{\sqrt{2}e^{(-\frac{1}{2}g)} \frac{\partial}{\partial r} h}{4(h^4 + 2h^3 - 2h - 1)} - \frac{\sqrt{2}\sqrt{h+1}\sqrt{-h+1} \frac{\partial}{\partial t} h}{2(h^4 + 2h^3 - 2h - 1)} \end{aligned}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = & \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} h \frac{\partial}{\partial r} h^2}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^4 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{32 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^3 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^5 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^4 \frac{\partial^2}{\partial r \partial t} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^3 \frac{\partial}{\partial r} h \frac{\partial}{\partial t} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^4 r^4 e^{(-g)} h \frac{\partial}{\partial r} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{3 R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^2 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^3 \frac{\partial^2}{\partial r \partial t} h}{rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r} \\
& - \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^2 \frac{\partial}{\partial r} h \frac{\partial}{\partial t} h}{rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r} + \frac{rh^8 \frac{\partial^2}{(\partial t)^2} g}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} \frac{\partial}{\partial r} h}{32 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{3 R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^2 r^3 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h \frac{\partial}{\partial r} h \frac{\partial}{\partial t} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \dots
\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}
\Phi_{11} = & -\frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{16(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^4 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{16(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h \frac{\partial}{\partial r} h^2}{16(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{64(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{32(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial}{\partial r} h^2}{64(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} h \frac{\partial}{\partial r} h}{16(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{64(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{rh^8 \frac{\partial^2}{(\partial t)^2} g}{8(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^4 e^{(-g)} \frac{\partial}{\partial r} h}{32(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{2(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{3rh^7 \frac{\partial^2}{(\partial t)^2} g}{4(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h \frac{\partial}{\partial r} h^2}{2(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{8(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{rh^6 \frac{\partial}{\partial t} g \frac{\partial}{\partial t} h}{8(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{rh^6 \frac{\partial}{\partial t} h^2}{2(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{rh^7 \frac{\partial^2}{(\partial t)^2} h}{2(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r e^{(-g)} h^4}{4(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{7rh^6 \frac{\partial^2}{(\partial t)^2} g}{4(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^2 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{8(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} \frac{\partial}{\partial r} h^2}{8(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{3rh^5 \frac{\partial}{\partial t} g \frac{\partial}{\partial t} h}{4(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{5rh^5 \frac{\partial}{\partial t} h^2}{2(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{5rh^6 \frac{\partial^2}{(\partial t)^2} h}{2(rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \dots
\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\begin{aligned}
\Phi_{22} = & \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} h \frac{\partial}{\partial r} h^2}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^4 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{32 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^3 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^5 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^4 \frac{\partial^2}{\partial r \partial t} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^3 \frac{\partial}{\partial r} h \frac{\partial}{\partial t} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^4 r^4 e^{(-g)} h \frac{\partial}{\partial r} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{3 R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^2 \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^3 \frac{\partial^2}{\partial r \partial t} h}{rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r} \\
& + \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h^2 \frac{\partial}{\partial r} h \frac{\partial}{\partial t} h}{rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r} + \frac{rh^8 \frac{\partial^2}{(\partial t)^2} g}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} \frac{\partial}{\partial r} h}{32 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{3 R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h \frac{\partial}{\partial t} g \frac{\partial}{\partial r} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^2 r^3 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^3 \sqrt{h+1} \sqrt{-h+1} e^{(-\frac{1}{2}g)} h \frac{\partial}{\partial r} h \frac{\partial}{\partial t} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \dots
\end{aligned}$$

$$\begin{aligned}
\Lambda = & -\frac{R^4 r^5 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{48 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{64 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} \frac{\partial}{\partial r} h^2}{192 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^5 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{48 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{rh^8 \frac{\partial}{\partial t} g^2}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{64 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{3rh^7 \frac{\partial}{\partial t} g^2}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{rh^8 \frac{\partial^2}{(\partial t)^2} g}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{6 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{rh^7 \frac{\partial}{\partial t} g \frac{\partial}{\partial t} h}{rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r} \\
& - \frac{7rh^6 \frac{\partial}{\partial t} g^2}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{3rh^7 \frac{\partial^2}{(\partial t)^2} g}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^2 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{6 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{39rh^6 \frac{\partial}{\partial t} g \frac{\partial}{\partial t} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{3rh^6 \frac{\partial}{\partial t} h^2}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{rh^7 \frac{\partial^2}{(\partial t)^2} h}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r e^{(-g)} h^4}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{7rh^5 \frac{\partial}{\partial t} g^2}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{7rh^6 \frac{\partial^2}{(\partial t)^2} g}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^2 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{8 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^3 e^{(-g)} \frac{\partial}{\partial r} h^2}{24 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{6 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{33rh^5 \frac{\partial}{\partial t} g \frac{\partial}{\partial t} h}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \dots
\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = & \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{48 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^4 r^4 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{3 R^4 r^5 e^{(-g)} h \frac{\partial}{\partial r} h^2}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^4 r^5 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{32 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{3 R^4 r^4 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{32 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{11 R^4 r^5 e^{(-g)} \frac{\partial}{\partial r} h^2}{96 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^4 r^5 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{48 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^4 r^4 e^{(-g)} h \frac{\partial}{\partial r} h}{16 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^4 r^5 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{32 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{3 R^4 r^4 e^{(-g)} \frac{\partial}{\partial r} h}{32 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} h^3 \frac{\partial^2}{(\partial r)^2} h}{6 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^2 r^2 e^{(-g)} h^3 \frac{\partial}{\partial r} h}{6 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{3 R^2 r^3 e^{(-g)} h \frac{\partial}{\partial r} h^2}{2 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^3 e^{(-g)} h^2 \frac{\partial^2}{(\partial r)^2} h}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^2 e^{(-g)} h^2 \frac{\partial}{\partial r} h}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{11 R^2 r^3 e^{(-g)} \frac{\partial}{\partial r} h^2}{12 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{R^2 r^3 e^{(-g)} h \frac{\partial^2}{(\partial r)^2} h}{6 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} + \frac{R^2 r^2 e^{(-g)} h \frac{\partial}{\partial r} h}{6 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& - \frac{R^2 r^3 e^{(-g)} \frac{\partial^2}{(\partial r)^2} h}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} - \frac{R^2 r^2 e^{(-g)} \frac{\partial}{\partial r} h}{4 (rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r)} \\
& + \frac{r e^{(-g)} h^2 \frac{\partial}{\partial r} h^2}{rh^8 + 4rh^7 + 4rh^6 - 4rh^5 - 10rh^4 - 4rh^3 + 4rh^2 + 4rh + r} - \dots
\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "